

Usitha Uduwakaarachchi, M.Sc.

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EDUCATION

- 2025 – present **Ph.D., Biomedical Engineering**, University of Texas Southwestern Medical Center, TX, USA
- 2023 – 2025 **M.Sc., Physics**, Binghamton University, NY, USA
- 2017 – 2021 **B.Sc., (Hons), Physics**, University of Colombo, Sri Lanka

RESEARCH EXPERIENCE

- 2025 – present **Graduate Student Researcher**, University of Texas Southwestern Medical Center, USA
Supervisor: Reto Fiolka. **Research:** *Adaptive Optics in Oblique Plane Microscopy*
- 2023 – 2025 **Graduate Researcher**, Department of Physics and Astronomy, Binghamton University, NY, USA
Supervisor: Lun Yu. **Research:** *High Harmonics Generation in Su-Schrieffer-Heeger (SSH) Chain*
- 2020 – 2021 **Undergraduate Researcher**, Department of Physics, University of Colombo, Sri Lanka
Supervisor: Siyath Gunawardane. **Research:** *Development of Schlieren Imaging Module to measure dynamic quantities in transparent fluids*

TEACHING EXPERIENCE

- 2023 – 2025 **Teaching Assistant, PHYS 121 General Physics 1 class (Fall 2023 & 2024), PHYS 122 General Physics 2 class (Spring 2024 & 2025), and PHYS 131 General Physics 1 (Calculus-Based) class (Summer 2024)**
Department of Physics and Astronomy, Binghamton University, Binghamton, NY, USA
- 2022 – 2023 **Teaching Assistant, Classical Physics course, Introduction to Quantum Mechanics course, and Electromagnetism course**
Faculty of Science, Sri Lanka Technological Campus, Sri Lanka
- 2021 – 2022 **Developer and Instructor, Physics Laboratory course and Electronic and Computing Laboratory course**
Department of Physics, University of Colombo, Sri Lanka
- 2021 – 2022 **Teaching Assistant, Physics of Semiconductor Devices and Electromagnetic Theory**
Department of Physics, University of Colombo, Sri Lanka

SKILLS

- Languages** English: Proficient, Sinhala: Native Language
- Programming** Fluent in Python, Proficient in C++, and MATLAB
- Engineering** Image Processing, Data Analysis, Numerical Analysis

AWARDS

- Finalists, ROBOFEST, the Annual Robotic Competition, Faculty of Engineering, Sri Lanka Institute of Information Technology, 2018
- Bronze medal, Sri Lankan Physics Olympiad Competition, Institute of Physics, Sri Lanka, 2015
- High Distinction, Australian National Chemistry Quiz (Senior Division), Royal Australian Chemical Institute, 2014
- High Distinction, Australian National Chemistry Quiz (Junior Division), Royal Australian Chemical Institute, 2012

LEADERSHIP/MEMBERSHIP

2019 – 2020	Junior Treasurer , Student's Union, Faculty of Science, University of Colombo
2017 – 2020	Member , Student's Union, Faculty of Science, University of Colombo
2017 – 2020	Member , Colombo Innovations and Robotics Club, University of Colombo
2018 – 2020	Member , Physics Society, University of Colombo
2017 – 2020	Member , Catholic Student's Movement, University of Colombo

REFERENCES

Available on request